

2) Wireshark

Áplique un filtro de visualización ... <Ctrl-/>						
No.	Time	Source	Destination	Protocol	Length	Info
6722	15.609196900	200.16.29.213	192.168.1.16	TCP	1466	443 → 56136 [ACK] Seq=12154252 Ack=10934 Win=55296 Len=1412 [TCP PDU reassembled in 6724]
6723	15.609198300	200.16.29.213	192.168.1.16	TCP	1466	443 → 56136 [ACK] Seq=12155664 Ack=10934 Win=55296 Len=1412 [TCP PDU reassembled in 6724]
6724	15.609212400	200.16.29.213	192.168.1.16	TLSv1.3	14174	Application Data, Application Data
6725	15.609214000	200.16.29.213	192.168.1.16	TCP	1466	443 → 56136 [ACK] Seq=12171196 Ack=10934 Win=55296 Len=1412 [TCP PDU reassembled in 6728]
6726	15.609214900	200.16.29.213	192.168.1.16	TCP	1466	443 → 56136 [ACK] Seq=12172608 Ack=10934 Win=55296 Len=1412 [TCP PDU reassembled in 6728]
6727	15.609215000	200.16.29.213	192.168.1.16	TCP	1466	443 → 56136 [ACK] Seq=12174020 Ack=10934 Win=55296 Len=1412 [TCP PDU reassembled in 6728]
6728	15.609235100	200.16.29.213	192.168.1.16	TLSv1.3	65006	Application Data, Application Data, Application Data, Application Data, Application Data, Application Data, Application Data, Application D
6729	15.609247200	200.16.29.213	192.168.1.16	TLSv1.3	26882	Application Data, Application Data, Application Data, Application Data
6730	15.609362800	192.168.1.16	200.16.29.213	TCP	54	56136 → 443 [ACK] Seq=10934 Ack=12267212 Win=4194048 Len=0
6731	15.609436100	200.16.29.213	192.168.1.16	TLSv1.3	1466	Application Data
6732	15.609459800	192.168.1.16	200.16.29.213	TCP	54	56136 → 443 [ACK] Seq=10934 Ack=12268624 Win=4192768 Len=0
6733	15.609921200	192.168.1.16	200.16.29.213	TCP	54	[TCP Window Update] 56136 → 443 [ACK] Seq=10934 Ack=12268624 Win=4194048 Len=0
6734	15.610377300	200.16.29.213	192.168.1.16	TLSv1.3	65006	Application Data, Application Data, Application Data, Application Data, Application Data, Application Data, Application Data, Application D
6735	15.610386400	200.16.29.213	192.168.1.16	TLSv1.3	2878	Application Data, Application Data
6736	15.610406500	192.168.1.16	200.16.29.213	TCP	54	56136 → 443 [ACK] Seq=10934 Ack=12336400 Win=4194048 Len=0
6737	15.611760000	200.16.29.213	192.168.1.16	TLSv1.3	7114	Application Data, Application Data, Application Data, Application Data
6738	15.611839400	192.168.1.16	200.16.29.213	TCP	54	56136 → 443 [ACK] Seq=10934 Ack=12343460 Win=4194048 Len=0
6739	15.612277700	200.16.29.213	192.168.1.16	TCP	1466	443 → 56136 [ACK] Seq=12343460 Ack=10934 Win=55296 Len=1412 [TCP PDU reassembled in 6740]
6740	15.612279500	200.16.29.213	192.168.1.16	TLSv1.3	1466	Application Data
6741	15.612300300	200.16.29.213	192.168.1.16	TLSv1.3	28294	Application Data, Application Data, Application Data, Application Data, Application Data, Application Data, Application Data, Application D
6742	15.612310300	200.16.29.213	192.168.1.16	TLSv1.3	1306	Application Data, Application Data
6743	15.612406500	192.168.1.16	200.16.29.213	TCP	54	56136 → 443 [ACK] Seq=10934 Ack=12375776 Win=4194048 Len=0
6744	15.703975700	192.168.1.9	192.168.1.16	TLSv1.2	302	Application Data, Application Data, Application Data
6745	15.703976900	192.168.1.9	192.168.1.16	TLSv1.2	302	Application Data, Application Data, Application Data
6746	15.948561900	192.168.1.16	142.251.155.2	UDP	661	55889 → 443 Len=619
6747	15.966507900	142.251.155.2	192.168.1.16	UDP	71	443 → 55889 Len=29
6748	15.975127100	192.168.1.16	142.251.155.2	UDP	75	55889 → 443 Len=33
6749	16.191246200	142.251.155.2	192.168.1.16	UDP	1188	443 → 55889 Len=1146
6750	16.192398800	142.251.155.2	192.168.1.16	UDP	97	443 → 55889 Len=55
6751	16.193771100	142.251.155.2	192.168.1.16	UDP	202	443 → 55889 Len=160
6752	16.195444500	192.168.1.16	142.251.155.2	UDP	78	55889 → 443 Len=36
6753	16.215126000	142.251.155.2	192.168.1.16	UDP	67	443 → 55889 Len=25
6754	16.238023700	192.168.1.16	192.168.1.2	TLSv1.2	164	Application Data
6755	16.238062400	192.168.1.16	192.168.1.2	TLSv1.2	164	Application Data
6756	16.238097400	192.168.1.16	192.168.1.2	TLSv1.2	164	Application Data
6757	16.348641500	192.168.1.2	192.168.1.16	TLSv1.2	164	Application Data
6758	16.349577900	192.168.1.2	192.168.1.16	TLSv1.2	164	Application Data
6759	16.349578500	192.168.1.2	192.168.1.16	TLSv1.2	164	Application Data
6760	16.402748700	192.168.1.16	192.168.1.2	TCP	54	53064 → 8009 [ACK] Seq=441 Ack=441 Win=251 Len=0
6761	16.402750000	192.168.1.16	192.168.1.2	TCP	54	55649 → 8009 [ACK] Seq=441 Ack=441 Win=254 Len=0
▶ Frame 6738: Packet, 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface \Device\NPF_{306D48D4-18B5-41FC-BD2B-25A99A2AE5A1}, id 0						
▶ Ethernet II, Src: CloudNetwork_60:91:bf (2c:98:11:60:91:bf), Dst: HuaweiTechno_25:41:f2 (14:8c:4a:25:41:f2)						
▼ Destination: HuaweiTechno_25:41:f2 (14:8c:4a:25:41:f2)						
.....0. = LG bit: Globally unique address (factory default)						
.....0. = IG bit: Individual address (unicast)						
▼ Source: CloudNetwork_60:91:bf (2c:98:11:60:91:bf)						
.....0. = LG bit: Globally unique address (factory default)						
.....0. = IG bit: Individual address (unicast)						
Type: IPv4 (0x0800)						
[Stream index: 4]						
▶ Ethernet II, Src: CloudNetwork_60:91:bf (2c:98:11:60:91:bf), Dst: HuaweiTechno_25:41:f2 (14:8c:4a:25:41:f2)						
▶ Internet Protocol Version 4, Src: 192.168.1.16, Dst: 200.16.29.213						
▶ Transmission Control Protocol, Src Port: 56136, Dst Port: 443, Seq: 10934, Ack: 12343460, Len: 0						
0000 14 8c 4a 25 41 f2 2c 98 11 60 91 bf 08 00 45 00 ...KA...E						
0010 00 28 15 e9 40 00 00 06 3d 49 c0 a8 01 10 c8 10 ...@...=I....						
0020 1d d5 db 48 01 bb 54 ab a9 e3 36 0c 0f 1c 50 10 ...H.T...6..P						
0030 3f ff a7 7c 00 00						

a) Direcciones MAC

Wireshark · Paquete 6738 · Wi-Fi	
▶ Frame 6738: Packet, 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface \Device\NPF_{306D48D4-18B5-41FC-BD2B-25A99A2AE5A1}, id 0	
▼ Ethernet II, Src: CloudNetwork_60:91:bf (2c:98:11:60:91:bf), Dst: HuaweiTechno_25:41:f2 (14:8c:4a:25:41:f2)	
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▼ Source: CloudNetwork_60:91:bf (2c:98:11:60:91:bf)	
.....0. = LG bit: Globally unique address (factory default)	
.....0. = IG bit: Individual address (unicast)	
Type: IPv4 (0x0800)	
[Stream index: 4]	

MAC Origen: 2c:98:11:60:91:bf (CloudNetwork_60:91:bf) → Placa de red de mi PC

MAC Destino: 14:8c:4a:25:41:f2 (HuaweiTechno_25:41:f2) → Router

Wireshark nos permite visualizarlos dado que los primeros 3 bytes de una dirección MAC identifican al fabricante, conocidos como OUI (Identificador Único de Organización). Así podemos ver que se trata de un paquete saliente, siendo el emisor físico local el adaptador de red de mi PC y el receptor local el router.

b) Direcciones IP

```
▼ Internet Protocol Version 4, Src: 192.168.1.16, Dst: 200.16.29.213
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  ▶ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
  Total Length: 40
  Identification: 0x15e9 (5609)
  ▶ 010. .... = Flags: 0x2, Don't fragment
  ...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 128
  Protocol: TCP (6)
  Header Checksum: 0x3d49 [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 192.168.1.16
  Destination Address: 200.16.29.213
  [Stream index: 20]
▶ Transmission Control Protocol, Src Port: 56136, Dst Port: 443, Seq: 10934, Ack: 12343460, Len: 0
```

IP Origen: 192.168.1.16 (Src) -> IPv4

IP Destino: 200.16.29.213 (Dst) -> IPv4

c) Comparación entre direcciones MAC e IP

Las direcciones MAC e IP no representan lo mismo

- Las **direcciones MAC** tienen alcance de tramo local (*hop-by-hop*)
- Las **direcciones IP** tienen alcance global extremo a extremo (*end-to-end*)

En este mismo ejemplo podemos observar como la IP de destino (200.16.29.213) identifica directamente al servidor web final en la red externa para viajar a través de múltiples puntos, mientras que la MAC solo identifica a mi router local.

d) Campo EtherType

```
▼ Ethernet II, Src: CloudNetwork_60:91:bf (2c:98:11:60:91:bf)
  ▶ Destination: HuaweiTechno_25:41:f2 (14:8c:4a:25:41:f2)
  ▶ Source: CloudNetwork_60:91:bf (2c:98:11:60:91:bf)
  Type: IPv4 (0x0800)
  [Stream index: 4]
```

Según lo que indica el campo EtherType, el protocolo que está encapsulado dentro de la trama es IPv4 (0x0800).